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EXP39:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

/* English letter frequencies (percent) A..Z */
double eng[26] = {8.17,1.49,2.78,4.25,12.70,2.23,2.02,6.09,6.97,0.15,0.77,4.03,2.41,
                  6.75,7.51,1.93,0.10,5.99,6.33,9.06,2.76,0.98,2.36,0.15,1.97,0.07};

typedef struct { int shift; double score; char text[500]; } Cand;

int main() {
    char ct[500];
    Cand c[26], tmp;
    int n, top, i, s, len;

    printf("Enter ciphertext: ");
    scanf("%s", ct);
    printf("How many top plaintexts do you want? ");
    scanf("%d", &top);
    len = strlen(ct);

    for (s = 0; s < 26; s++) {
        int cnt[26] = {0};
        double chi = 0.0;
        for (i = 0; i < len; i++) {
            int p = (toupper(ct[i]) - 'A' - s + 26) % 26;
            c[s].text[i] = p + 'a';
            cnt[p]++;
```

```
        }
        c[s].text[len] = '\0';
        for (i = 0; i < 26; i++) {
            double expct = eng[i] * len / 100.0;
            chi += (cnt[i] - expct) * (cnt[i] - expct) / expct;
        }
        c[s].shift = s;
        c[s].score = chi;           /* smaller chi-square = closer to English */
    }

    /* sort by likelihood (smallest chi-square first) */
    for (i = 0; i < 25; i++)
        for (s = 0; s < 25 - i; s++)
            if (c[s].score > c[s + 1].score) { tmp = c[s]; c[s] = c[s + 1]; c[s + 1] = tmp; }

    if (top > 26) top = 26;
    printf("\nRank  Key  Chi-square  Plaintext\n");
    for (n = 0; n < top; n++)
        printf("%3d   %2d   %8.2f    %s\n", n + 1, c[n].shift, c[n].score, c[n].text);
    return 0;
}
```

OUTPUT:

```
Enter ciphertext: HPLCPXPPETYRLEESPWTMCLCJEZXZCCZHLQEPCESPPILXLYOHTWWOTDNFDDESPACZUPNEHZCVTYOPE
LTW
How many top plaintexts do you want? 3

Rank  Key  Chi-square  Plaintext
  1   11      24.46    wearemeetingatthelibrarytomorrowaftertheexamandwilldiscusstheprojectwork
indetail
  2    1     272.32    gokbowoodsxqkddrovslbkbidywybbygkpdobdroohkwkxngsvvnscmeccdrozbytomdgybu
sxnodksv
  3    4     314.77    dlhyltllapunhaaolspiyhyfavtvyyvdhmalyaollehthukdpsskpzjbzzaolwyvqljadvyr
puklahps
```